

Magnetic and Ferroelectric Manipulation of Valley Physics in Janus Piezoelectric Materials

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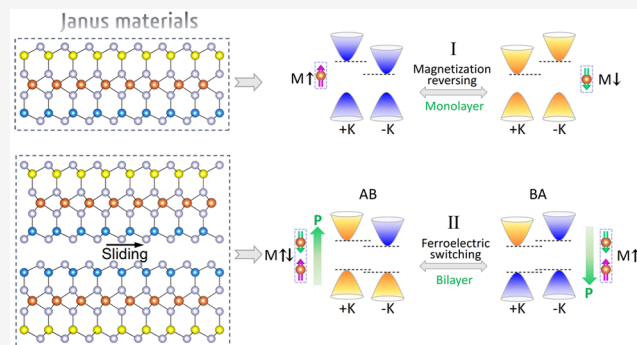
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ABSTRACT: The realization of multiferroic materials offers the possibility of multifunctional electronic device design. However, the coupling between the multiferroicity and piezoelectricity in Janus materials is rarely reported. In this study, we propose a mechanism for manipulating valley physics by magnetization reversing and ferroelectric switching in multiferroic and piezoelectric material. The ferromagnetic VSiGeP₄ monolayer exhibits a large valley polarization up to 100 meV, which can be effectively operated by reversing magnetization. Interestingly, the antiferromagnetic VSiGeP₄ bilayers with AB and BA stacking configurations allow the coexistence of valley polarization and ferroelectricity, supporting the proposed strategy for manipulating valley physics via ferroelectric switching and interlayer sliding. In addition, the VSiGeP₄ monolayer contains remarkable tunable piezoelectricity regulated by electron correlation U . This study proposes a feasible idea for regulating valley polarization and a general design idea for multifunctional devices with multiferroic and piezoelectric properties, facilitating the miniaturization and integration of nanodevices.

KEYWORDS: Valley polarization, Ferromagnetic, Antiferromagnetic, Ferroelectricity, Piezoelectricity, Janus materials



Two-dimensional (2D) materials attract widespread attention due to their remarkable and compelling physical and chemical properties, and they hold great potential for the development of next-generation electronic devices.^{1–7} In addition to charge and spin, the valley is a new degree of freedom in 2D materials, which has broad applications in valleytronics such as encoding and storing information. The valley is represented by the local energy extremum of the conduction band (CB) or valence band (VB).^{8–15} Generally, due to the inherent magnetic exchange interaction and spatial inversion symmetry breaking, the valley polarization (VP) is spontaneously introduced in 2D ferromagnetic systems after considering spin–orbit coupling (SOC).^{16,17} Valley polarization is prone to occur in several 2D single-layer materials, such as the magnetic VSe₂,¹⁶ VSSe,¹⁸ AuCl₃,¹⁹ CeI₂,²⁰ RuCl₂,²¹ RuClX (X = F, Br),²² VSi₂P₄,^{23,24} and VGe₂P₄ monolayers.²⁵ It is noteworthy that there is very little research on whether the VP generates in 2D multilayer materials. In addition, whether the VP can be effectively modulated by the stacking modes in the multilayer is worthy studying.

Previous studies reported that the stacking configurations of multilayers can affect its physical properties,^{26–28} such as the direction of polarization. The sliding ferroelectricity (FE) via interlayer sliding occurs in BN and AlN bilayers, Graphene/BN and SnS/SnSSe heterostructures.^{29,30} And the novel FE

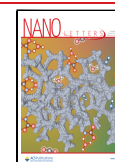
properties in a BN bilayer^{31,32} and transition metal dichalcogenides (TMD) heterobilayers³³ have been demonstrated experimentally. Additionally, stacking orders effectively adjust the electronic state structure, leading to the layer-polarized anomalous Hall effect (LP-AHE), such as the MnBi₂Te₄³⁴ and VSi₂P₄ bilayers.³⁵ Interestingly, stacking order can regulate the intrinsic VP of ferrovalley (FV) materials, favoring the multiferroics behaviors in YI₂, LaI₂ and GdBr₂ bilayers.³⁶

Recently, the MA₂Z₄ family with the Z-A-Z-M-Z-A-Z seven-layer atomic structure have been successfully synthesized experimentally.³⁷ For example, the MoSi₂N₄ monolayer with $\alpha 1$ phase can be constructed by embedding a MoN₂ layer into two Si–N bilayers.³⁷ Researchers also found that this family of materials possesses the emerging magnetic, valley and topological characteristics.^{38–41} For example, due to the broken inversion and time-reversal symmetries, the $\alpha 1$ phase VSi₂P₄ monolayer has the ferromagnetic semiconductor

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properties and valley physics.²⁴ The VGe_2P_4 monolayer with $\alpha 2$ phase is a quasi-half-valley metal, and only one type of carriers exhibits the VP.²⁵

The broken central symmetry endows 2D Janus materials with piezoelectric properties, serving as urgently sought candidates for advanced equipment in a microelectromechanical system (MEMS).^{42,43} The coexistence of multi-ferroic and piezoelectric properties promotes the miniaturization and integration of nanodevices. Herein, in this work, we focus on the intriguing physic properties caused by symmetry breaking in the novel VSiGeP_4 monolayer with $\alpha 1$ phase and the effect of stacking orders on the VP of VSiGeP_4 bilayers. We theoretically reported that the VSiGeP_4 monolayer exhibits a large VP, which is manipulated by reversing magnetization. Surprisingly, VSiGeP_4 bilayers endow the coexistence of VP and FE, supporting the proposed strategy for manipulating VP via ferroelectric switching by interlayer sliding. Moreover, the VSiGeP_4 monolayer contains remarkable in-plane and out-of-plane piezoelectric properties, which are regulated by the electron correlation U . The operation mechanism for VP in Janus materials is shown in Figure 1. In the Janus piezoelectric

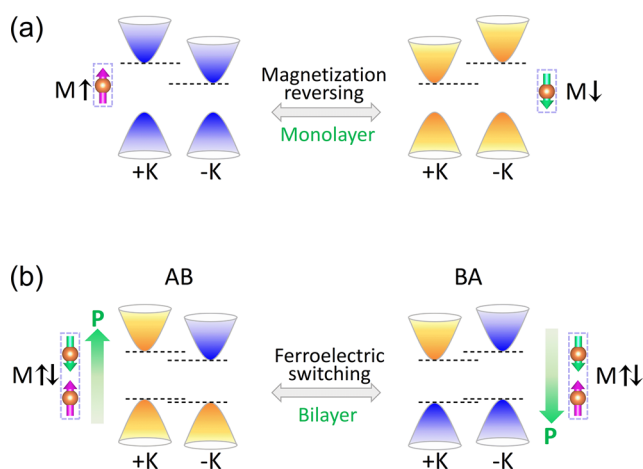


Figure 1. Operation mechanisms for valley polarization by magnetization reversing and ferroelectric switching in the monolayer (a) and bilayer (b). Orange and blue cones correspond to spin-down and spin-up bands. Green arrows represent ferroelectric polarization P .

monolayer, the VP can be operated by the reversing the magnetic moment of metal atoms. In antiferromagnetic bilayers, the ferroelectric switching manipulates the VP. The AB stacking state transforms to the BA stacking state via interlayer sliding without changing of magnetization, and the ferroelectric polarization (remarked by green arrows) is switched from the $+z$ axis direction to the $-z$ axis direction.

The fully optimized VSiGeP_4 monolayer contains a P-Si-P-V-P-Ge-P seven-layer atomic structure (in Figure 2a). The VSiGeP_4 Janus structure is a sandwich structure, which can be regarded that the middle VP_2 layer is sandwiched by top SiP and bottom GeP layers. It is a member of the MA_2X_4 family and has the possibility of being successfully prepared by chemical vapor deposition (CVD) similar to MoSi_2N_4 film.³⁷ Mirror symmetry is breaking with respect to the middle VP_2 layer, thus the symmetry of VSiGeP_4 (No. 156) is lower than that of VSi_2P_4 ²³ and VGe_2P_4 ²⁵ (No. 187). A unit cell of the VSiGeP_4 monolayer consists of one V, Si, and Ge and four P atoms, forming a honeycomb hexagonal honeycomb structure. Using the rectangle supercell of VSiGeP_4 monolayer, the

energy difference $\Delta E = E_{\text{AFM}} - E_{\text{FM}}$ between antiferromagnetic (AFM) and ferromagnetic (FM) orderings (in Figure S1) is found to be 40.68 meV, indicating that VSiGeP_4 contains the FM ground state. The energy difference between in-plane and out-of-plane magnetization is regarded as the magnetic anisotropy energy (MAE), determining the stability of FM order in monolayers. From the Figure S2, the MAE for VSiGeP_4 monolayer equals to 51.54 $\mu\text{eV}/\text{f.u.}$ and is isotropic in the xy plane and highly depends on the magnetization direction in xz and yz planes, suggesting that VSiGeP_4 contains the out-of-plane easy axis. Calculation predicts a magnetic moment of 1 μB per unit cell for V atom in the VSiGeP_4 monolayer. The lattice constant a for the fully optimized VSiGeP_4 is 3.557 Å, which is between ones of VSi_2P_4 (3.486 Å)²³ or VGe_2P_4 (3.560 Å)²⁵ monolayers. According to the analysis of the ELF diagrams (as shown in Figure 2b), the Si and Ge atoms are covalently bonded to two adjacent P atoms, which is confirmed by the localized charge distribution between them. However, the V atoms form the ionic bonds with P atoms, which are verified by the highly localized around them. According to the results of *ab initio* molecular dynamics (AIMD) simulation at a high temperature of 500 K (in Figure S3), the energy of each atom fluctuates in a small range and the structural skeleton can maintain its integrity after the AIMD simulation, indicating that the VSiGeP_4 monolayer contains excellent thermal stability. In addition, its dynamic stability is verified by the phonon band structure (in Figure 2c). There is no appreciable imaginary frequency in the first BZ along the high-symmetry line. Noteworthy, according to the projected phonon density of states (PhDOS), the maximum frequency in the phonon spectrum is contributed by the hybridization of Si and P atoms. Those results confirm that the mechanical robustness of the covalent Si–P bonds. In addition, the elastic stiffness coefficients C_{kj} ($C_{11} = C_{22} = 179.49$ N/m, $C_{12} = 49.15$ N/m, $C_{66} = 65.17$ N/m) of VSiGeP_4 monolayer meet the Born–Huang mechanical stability rules:^{44,45} $(C_{11}C_{22} - C_{12}^2) > 0$ and $C_{66} > 0$, suggesting its greatly mechanical stability.

From the Figure 2d, VSiGeP_4 monolayer is an in-direct semiconductor with band gap of 0.10 eV. The conduction band minimum (CBM) for VSiGeP_4 is located at the high symmetry of $+K$ and $-K$ points and contributed by the spin-up channel, forming the degenerate valleys. While the valence band maximum (VBM) is located at the high symmetry Γ point, which is contributed by the spin-down channel. It is noteworthy that VSiGeP_4 exhibit the degenerate valleys in valence band (VB) as well due to the local extrema in VB at the $+K$ and $-K$ point. The VSiGeP_4 monolayer is a bipolar ferromagnetic semiconductor. The indirect bandgap semiconductor character of the VSiGeP_4 monolayer can hinder the free charge transfer between the VB and CB, endowing the preservation of polarized charge secondary retention and maintenance of spontaneous polarization. The V atom has a valence electron configuration of $3d^34s^2$. Importantly, the V atom lies in a trigonal prismatic crystal field environment in the VSiGeP_4 monolayer, which donates four electrons to neighboring P atoms. And its d orbital splits into the d_z^2 , $d_{xy} + d_{x^2-y^2}$ and $d_{yz} + d_{zx}$ orbitals. As shown in Figures 2e and S4, the CBM is provided by the $d_{xy} + d_{x^2-y^2}$ orbitals, the VBM is provided by the $d_{yz} + d_{zx}$ orbitals. While the local extrema in VB is provided by the d_z^2 orbital, which is occupied by only one electron. Therefore, a magnetic moment for each V atom

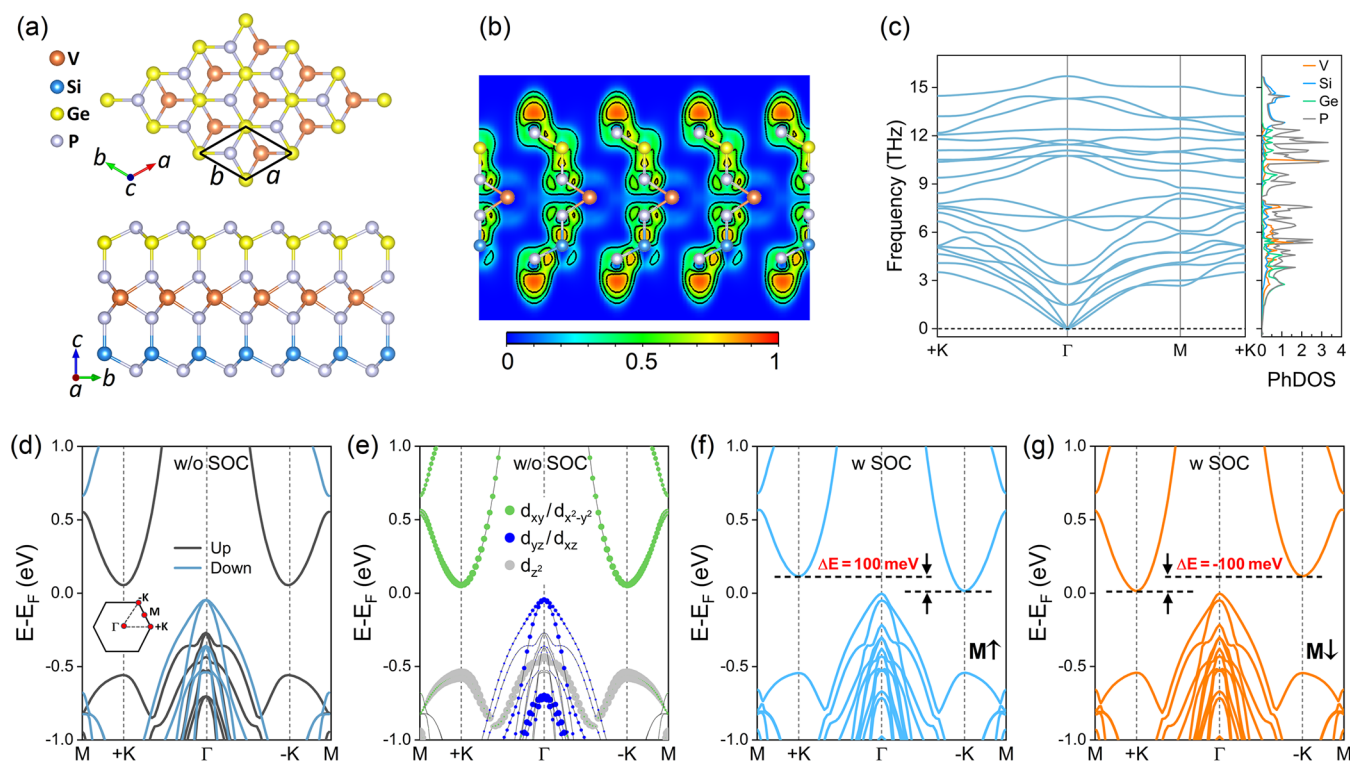


Figure 2. (a) Crystal structure, (b) electron localization function (ELF) map, (c) phonon spectrum with projected phonon density of states (PhDOS), (d,e) spin-polarized and orbital-resolved band structures without SOC, and (f,g) band structures with SOC for the VSIGeP₄ monolayer with magnetic moment of V atom pointing to +z and -z axis directions, respectively. Inset in (d) is the first Brillouin zone (BZ). Fermi level $E_F = 0$.

is expected to be 1.0 μ_B , which is greatly consistent with the calculated value.

The intriguing VP is introduced when the SOC is included; see Figure 2f,g. In the CB, the valley at +K point moves upward, while the valley at -K point moves downward, leading to the large valley splitting $\Delta E = 100$ meV. While for the VB, the valley splitting at +K and -K points can be ignored. In contrast, when the magnetization direction of V atoms is reversed, the valley splitting ΔE is equal to -100 meV in CB. In other words, the valley at -K point shifts upward, which is higher than that at +K point. It is concluded that reversing the magnetization of the V atom in VSIGeP₄ is a useful strategy for switching the VP. The gap value of VSIGeP₄ is about 17 meV with considering the SOC. In the following, we use a simple model^{24,25} to describe the SOC term to get the basic physics of VP in VSIGeP₄:

$$\hat{H}_{\text{SOC}} = \lambda \hat{L} \cdot \hat{S} = \hat{H}_{\text{SOC}}^0 + \hat{H}_{\text{SOC}}^1 \quad (1)$$

where, the orbital angular momentum $\hat{L}$, the spin angular momentum $\hat{S}$, the interactions between the same spin states $\hat{H}_{\text{SOC}}^0$ and the interactions between opposite spin states $\hat{H}_{\text{SOC}}^1$. According to the Figure 2d, both the +K and -K valleys in the CB and VB are contributed by independent different spin channels, thus the interaction term $\hat{H}_{\text{SOC}}^1$ can be neglected here. For the $\hat{H}_{\text{SOC}}^0$, it can be expressed by the θ and ϕ :^{24,25}

$$\hat{H}_{\text{SOC}}^0 = \lambda \hat{S}_z \left(\hat{L}_z \cos \theta + \frac{1}{2} \hat{L}_+ e^{-i\phi} \sin \theta + \frac{1}{2} \hat{L}_- e^{+i\phi} \sin \theta \right) \quad (2)$$

The VSIGeP₄ contain the out-of-plane magnetization ($\theta = 0$ or π); thus, the $\hat{H}_{\text{SOC}}^0$ is simplified as

$$\hat{H}_{\text{SOC}}^0 = \lambda \hat{S}_z \hat{L}_z = \alpha \hat{L}_z \quad (3)$$

where $\hat{L}_z$ is the orbital angular momentum along the z direction and α is the coupling strength.

Considering the contribution of orbital and the wave vector symmetry at the +K and -K valleys, basis functions are chosen as follows:

$$|\varphi_c^\tau\rangle = \sqrt{\frac{1}{2}} (|d_{x^2-y^2}\rangle + i\tau |d_{xy}\rangle) \quad (4)$$

$$|\varphi_v^\tau\rangle = |d_{z^2}\rangle \quad (5)$$

where the subscripts c, v, and τ represent CB, VB, and valley index ($\tau = \pm 1$). Then, the energy of +K and -K valleys in CB and VB can be written as

$$E_c^\tau = \langle \varphi_c^\tau | \hat{H}_{\text{SOC}}^0 | \varphi_c^\tau \rangle \quad (6)$$

$$E_v^\tau = \langle \varphi_v^\tau | \hat{H}_{\text{SOC}}^0 | \varphi_v^\tau \rangle \quad (7)$$

Considering the distribution of the d orbitals of V atoms, the valley splitting in the CB and VB is expressed as

$$|\Delta E^c| = E_c^K - E_c^{-K} = i \langle d_{x^2-y^2} | \hat{H}_{\text{SOC}}^0 | d_{xy} \rangle - i \langle d_{xy} | \hat{H}_{\text{SOC}}^0 | d_{x^2-y^2} \rangle = 4\alpha \quad (8)$$

$$|\Delta E^v| = E_v^K - E_v^{-K} = 0 \quad (9)$$

Here, we have made use of:

$$\hat{L}_z |d_{x^2-y^2}\rangle = 2i |d_{xy}\rangle \quad (10)$$

$$\hat{L}_z |d_{xy}\rangle = -2i |d_{x^2-y^2}\rangle \quad (11)$$

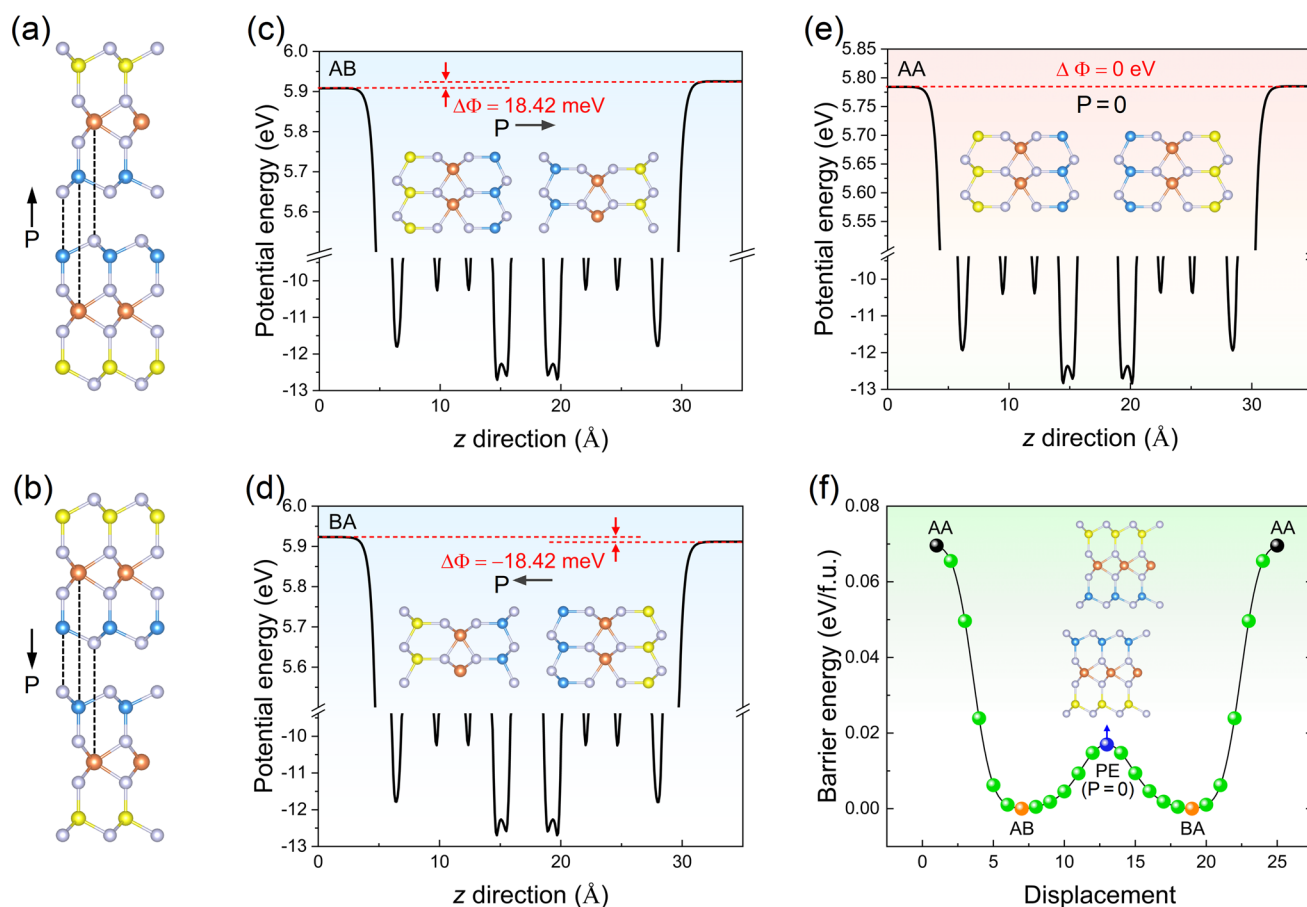


Figure 3. (a,b) AB and BA stacking states of the AFM VSiGeP₄ bilayer. (c–e) Planar-average electrostatic potentials along the *z*-axis direction for AB, BA and AA stacking states. (f) Ferroelectric switching pathways and energy barriers via interlayer sliding in VSiGeP₄ bilayers. Ferroelectric polarization *P* is indicated by black arrows.

Clearly, the analysis is consistent with the results presented in Figure 2f,g. That is, a large spontaneous VP appears in the CB, but not in the VB.

The VSiGeP₄ bilayer with the AA, AB, BA, AA', AB' and BA' stacking configurations are constructed (see Figure S5). The AB and BA structures are both ground states with the lowest total energy, and six stacking structures have the AFM ordering (in Figure S6a). Except the AB' and BA', the other four structures possess the out-of-plane easy axis (in Figure S6b), contributing to get spontaneous VP. Here, we mainly focus on the AB and BA stacking states. For the AB (BA) stacking bilayer (in Figure 3a,b), the Si (V) atom of the upper layer is located directly above the V (Si) atom of the lower layer, the V (P) atom of the upper layer sits directly above the P (V) atom of the lower layer, and the P (Si) of the upper layer is right above the Si (P) of the lower layer, resulting in the out-of-plane electric polarization *P* produced in the AB (BA) stacking bilayer. As shown in Figure 3c,d, the positive (negative) discontinuity $\Delta\Phi = 18.42$ (−18.42) meV between the vacuum levels of upper and lower layers clearly verifies the spontaneous *P* pointing upward (downward) in the AB (BA) stacking bilayer. The lattice constant *a* of the fully relaxed VSiGeP₄ bilayer with AB and BA stacking configurations is 3.53 Å, and their interlayer distance *d* is 3.19 Å. The AB and BA stacking bilayers can realize mutual transformation through interlayer sliding, so we can conclude that the sliding FE exists in VSiGeP₄ bilayer, which is also demonstrated from the general theory for bilayer stacking ferroelectricity.⁴⁶ Due to

charge transfer between the upper and lower layers, the AB (BA) state exhibits strong out-of-plane piezoelectric polarization of 0.87×10^{-10} (-0.87×10^{-10}) C/m. Based on the berry phase method,^{47,48} the ferroelectric polarization for AB (BA) state is 2.01×10^{-12} C/m, which is comparable to that of BN (2.08×10^{-12} C/m)²⁹ and MoSi₂N₄ (3.36×10^{-12} C/m)⁴⁹ bilayers.

There are two ways to switch polarization in VSiGeP₄ bilayer: (i) the AB stacking state transitions to AA stacking with mirror symmetry (in Figure 3e), and then to BA stacking configuration. (ii) the AB stacking state is directly transformed into the BA stacking state through the intermediate paraelectric phase (PE) state. From Figure 3f, it is found that the FE switching energy barrier through the first (i) way is 0.069 eV/f.u., while that through the second (ii) way is only 0.017 eV/f.u., while that through the second (ii) way is only 0.017 eV/f.u. Therefore, the FE switching between AB and BA stacked bilayer VSiGeP₄ tends to be achieved in the (ii) way. These switching barriers are larger than that of 0.0006 eV/f.u. for the WTe₂ bilayer⁵⁰ but much lower than that of 0.6 eV/f.u. for the 1T'-CrCoS₄,⁵¹ suggesting that the room-temperature polarization switching in VSiGeP₄ bilayers is easy to be implemented.

From Figure 4a–d, the AB and BA stacking states are indirect semiconductors with in-direct band gaps of 16 meV. The CBM is located at the high symmetry of the −K (+K) point and is contributed by the spin-up (spin-down) channel for the AB (BA) stacking state. The VBM for AB (BA) stacking state is located at the high symmetry Γ point and is contributed by the

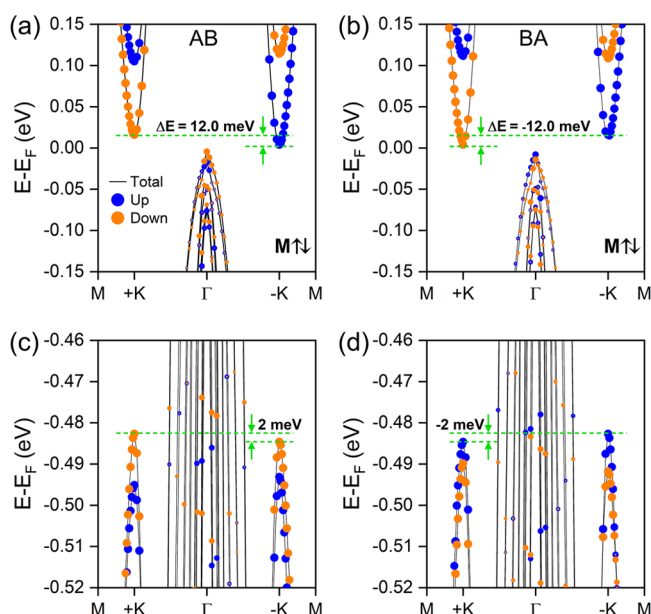


Figure 4. Spin-polarized band structures with SOC for VSiGeP₄ bilayer with the AB (a,c) and BA (b,d) stacking configurations. Fermi level $E_F = 0$.

spin-down (spin-up) channel. The local extrema of valence bands at the +K and -K points for AB (BA) stacking state are

contributed by the spin-down (spin-up) channel. Obviously, the intriguing valley polarization ΔE of 12 (−12) meV can be induced in the CB for AB (BA) stacking state. While in the VB, the valley splitting is only to be 2 (−2) meV in the AB (BA) stacking state. Therefore, the VP is reversed by the ferroelectric switching via interlayer sliding in VSiGeP₄ bilayers, which provides a feasible strategy for valley electronics regulation via controlling stacking modes. Band structures for other stacking structures are shown in Figure S7. Large VP occurs at CB and can only be switched by reversing magnetization. As shown in Figure S8, when the magnetic moment of V atom of upper (lower) layer in the BA' bilayer switches from +z (−z) to −z (+z) direction, the VP switches from 97 to −97 meV.

VSiGeP₄ monolayer exhibits great flexible mechanical properties with a smaller Young's modulus $Y(\theta)$ of 166.03 N/m and a larger Poisson's ratio $\nu(\theta)$ of 0.27 (in Figure S9). Broken inversion symmetry allows uneven charge distribution and benefits the introduction of spontaneous piezoelectricity. It is demonstrated by the planar-average charge density and planar-average electrostatic potential with discontinuities of $\Delta\Phi = 33.86$ meV (in Figure 5a). According to the details of piezoelectric stress (e_{ij}) and strain (d_{ik}) coefficients (in the Supporting Information), the VSiGeP₄ not only possesses the in-plane polarization of $e_{22} = 1.11 \times 10^{-10}$ C/m ($d_{22} = 0.87$ pm/V), but also the rare out-of-plane polarization of $e_{31} = e_{32} = 0.25 \times 10^{-10}$ C/m ($d_{31} = d_{32} = 0.09$ pm/V) and $e_{33} = 0.35 \times 10^{-10}$ C/m ($d_{33} = 3.73$ pm/V), which are greatly larger than that of many reported materials.^{18,52,53} These results show that

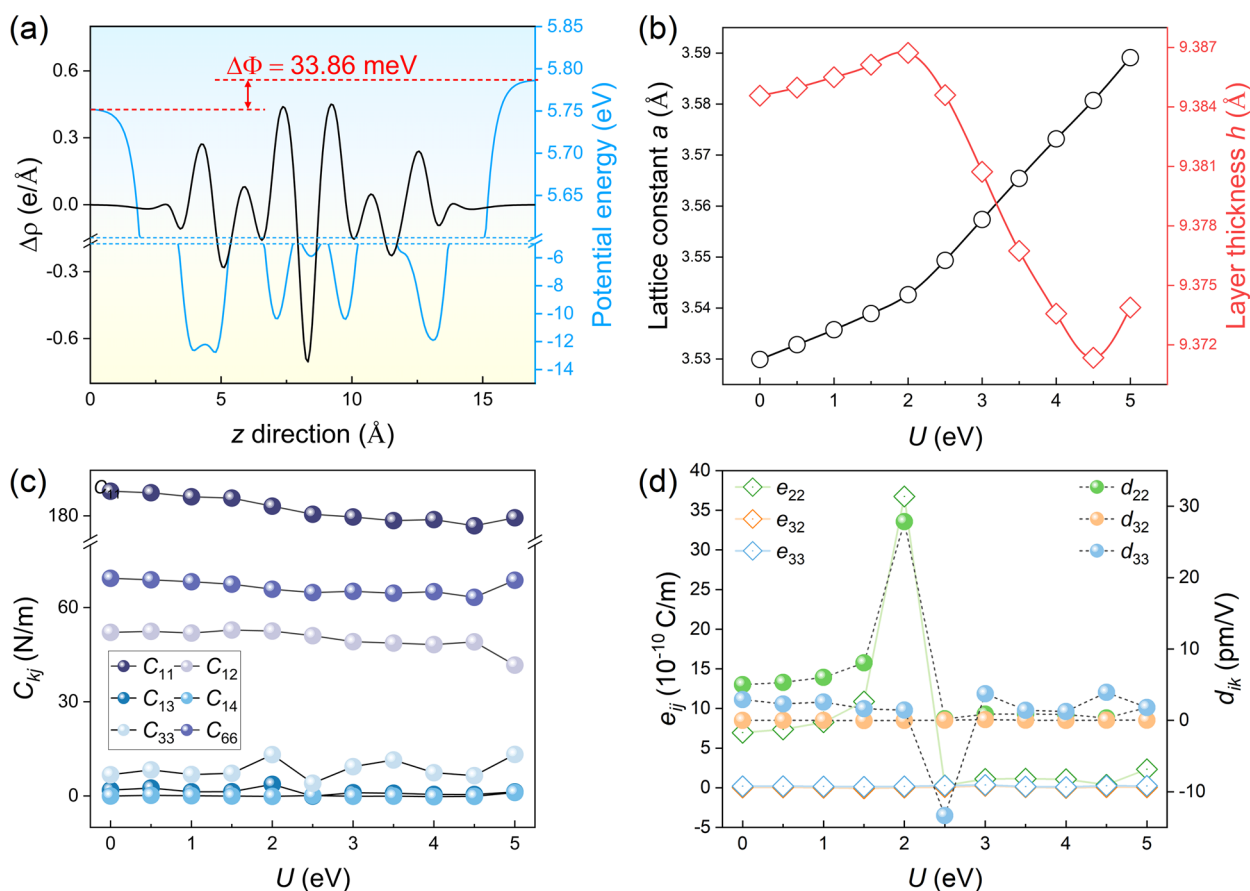


Figure 5. (a) Planar-average charge density and planar-average electrostatic potential along the z-axis direction for the VSiGeP₄ monolayer. (b) Lattice constants a and layer thickness h , (c) relaxed-ion elastic coefficients C_{kj} and (d) piezoelectric stress e_{ij} and strain d_{ik} coefficients as a function of U for the VSiGeP₄ monolayer.

VSiGeP₄ can serve as the long-sought candidate material for piezoelectric devices with an excellent piezoelectric response. In addition, we find that electron correlation ($U = 0\text{--}5.0$ eV) can regulate electronic and piezoelectric properties of the Janus VSiGeP₄ monolayer. The VSiGeP₄ always keeps the semiconductor characteristics with different $U = 0\text{--}5.0$ eV (in Figure S10). The elastic constants C_{kj} (in Figure 5c) of VSiGeP₄ monolayer changes with U , but the dependence on U is weak. As shown in Figure 5d, e_{ij} and d_{ik} depend on the value of U . The changes of e_{32} and e_{33} for VSiGeP₄ are not obvious with the increase of U . However, e_{22} varies significantly with U and reaches a maximum value of 36.74×10^{-10} C/m at $U = 2.0$ eV, which origins from the increases of lattice constant a and the maximum h of 9.386 Å (in Figure 5b). Correspondingly, the d_{22} is also highly dependent on U and reaches a maximum value of 27.84 pm/V at $U = 2.0$ eV. Therefore, manipulating the intensity of correlation strength determined by the Hubbard U is a useful strategy to regulate electronic and piezoelectric properties of 2D materials. The VSiGeP₄ structure is a potential candidate for designing multifunctional nanodevices that require both multiferroic and piezoelectric properties. Designing materials with this similar structure are expected to achieve multiple functional coupling, which is conducive to the miniaturization and integration of nanodevices.

CONCLUSION

In summary, we propose a mechanism for manipulating valley physics by magnetization reversing and ferroelectric switching in multiferroic and piezoelectric materials. The mechanism is demonstrated in the VSiGeP₄ monolayer and bilayers. The ferromagnetic VSiGeP₄ monolayer exhibits a large VP of 100 meV, which can be effectively operated by reversing magnetization. Interestingly, the coexistence of valley and ferroelectric polarizations is realized in VSiGeP₄ bilayers with antiferromagnetic stacking states. The VP can be reversed by controlling the direction of ferroelectric polarization by interlayer sliding in the AB and BA stacking states. In addition, the VSiGeP₄ monolayer contains remarkable in-plane and out-of-plane piezoelectric properties, which can be regulated by electron correlation U . This study proposes a feasible idea for the regulation of valley polarization and a general design idea for multifunctional devices with a multiferroic property and piezoelectric response, supporting the miniaturization and integration of nanodevices.

ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge at <https://pubs.acs.org/doi/10.1021/acs.nanolett.3c03238>.

Computational methods, details of piezoelectric stress e_{ij} and strain d_{ik} coefficients. Rectangle supercell FM and AFM configurations, angular dependence of MAE, AIMD simulation, orbital-resolved band structures, Young's modulus $Y(\theta)$ and Poisson's ratio $\nu(\theta)$, band structures at different U for VSiGeP₄ monolayer. Crystal stacking structures, energy difference $\Delta E'$ between AFM and FM orderings and the MAE, band structures of AA', AB' and BA' stacking states for VSiGeP₄ bilayers (PDF)

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Notes

The authors declare no competing financial interest.

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